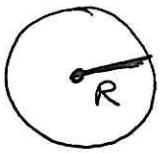
P4.10



$$\vec{P}(\vec{r}) = k\vec{r}$$

(a) $\rho_b = -\vec{\nabla}\cdot\vec{P} = -k\,\vec{\nabla}\cdot\vec{r} = -3k$

as $\vec{\nabla}\cdot\vec{r} = \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right)\cdot\left(x\hat{x} + y\hat{y} + z\hat{z}\right) = 3$

$$\sigma_b = \vec{P}\cdot\hat{r}\Big|_{r=R} = kR$$

(b) Electric field $\vec{E} = \vec{E}_\rho + \vec{E}_\sigma$

$\vec{E}_\rho$: field of a uniformly charged sphere, $\rho_b = -3k$

$\vec{E}_\sigma$: field of spherical shell $\sigma_b = kR$

Inside the sphere: $r<R$

$\vec{E}_\sigma(r<R) = 0$ by Gauss's Theorem.

$$\int \vec{E}_\rho\cdot d\vec{S} = \frac{-3k\cdot\frac{4\pi}{3}r^3}{\varepsilon_0} \Rightarrow E_\rho\cdot 4\pi r^2 = -\frac{4\pi k}{\varepsilon_0}r^3$$

$$\vec{E}_\rho(r<R) = -\frac{k\vec{r}}{\varepsilon_0}$$

$$\vec{E}(r) = -\frac{k}{\varepsilon_0}\vec{r} \quad \text{for } r<R.$$

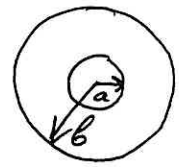
Outside the sphere: $r>R$

$$\left.\begin{aligned} \vec{E}_\sigma &= \frac{\sigma_b\cdot 4\pi R^2}{4\pi\varepsilon_0 r^2}\hat{r} = \frac{R^3 k\,\hat{r}}{\varepsilon_0 r^2} \\ \vec{E}_\rho &= \frac{\rho_b\cdot\frac{4\pi}{3}R^3}{4\pi\varepsilon_0 r^2}\hat{r} = -\frac{R^3 k\,\hat{r}}{\varepsilon_0 r^2} \end{aligned}\right\} \vec{E}(r) = 0 \text{ for } r>R.$$

Thus
$$\left.\begin{aligned} \vec{D}(r<R) &= \varepsilon_0\vec{E} + \vec{P} = -k\vec{r} + k\vec{r} = 0 \\ \vec{D}(r>R) &= \varepsilon_0\vec{E} + \underbrace{\vec{P}}_{=0} = 0 \end{aligned}\right\} \begin{aligned} &D_r(r=R-0) = \\ &= D_r(r=R+0) \end{aligned}$$

as it should be.

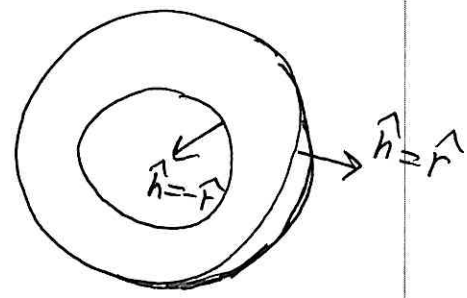
P4.15 $\vec{P}(\vec{r}) = \frac{k}{r}\hat{r}$



(a) $\rho_b = -\vec{\nabla}\cdot\vec{P} = -\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \cdot \frac{k}{r}\right) = -\frac{k}{r^2}$

$\sigma_b(r=a) = \vec{P}\cdot(-\hat{r})\Big|_{r=a}$

$= -\frac{k}{a}$



$\sigma_b(r=b) = \vec{P}\cdot\hat{r}\Big|_{r=b} = \frac{k}{b}$

<u>$r<a$</u> $\vec{E}=0$ because no charge inside.

<u>$a<r<b$</u> $\vec{E}(\vec{r}) = \vec{E}_a(\vec{r}) + \vec{E}_\rho(\vec{r})$

field of shell of radius a

field of charge distribution $\rho = -\frac{k}{r^2}$

Note that $\vec{E}_b(\vec{r})$ = field of shell of radius b does not contribute for <u>$r<b$</u>.

$$\vec{E}_a(\vec{r}) = \frac{\sigma(r=a)\cdot 4\pi a^2}{4\pi\varepsilon_0 r^2}\hat{r} = \frac{-ka}{\varepsilon_0 r^2}\hat{r}.$$

Via Gauss's law $\vec{E}_\rho(\vec{r})$: $E_\rho(r)\cdot 4\pi r^2 = \frac{1}{\varepsilon_0}\int_a^r \rho_b \cdot 4\pi r^2 dr =$

$$= -\frac{4\pi k}{\varepsilon_0}(r-a)$$

$$\vec{E}_\rho(\vec{r}) = -\frac{k(r-a)}{\varepsilon_0 r^2}\hat{r}.$$

$\Rightarrow \vec{E}(\vec{r}) = -\frac{kr+ka-ka}{\varepsilon_0 r^2}\hat{r} = -\frac{k\hat{r}}{\varepsilon_0 r}$ inside the shell.

$\underline{r>b}$ $\quad \vec{E}(\vec{r}) = \vec{E}_a(\vec{r}) + \vec{E}_\rho(\vec{r}) + \vec{E}_b(\vec{r})$

field of outer shell of radius $\underline{b}$.

$$\vec{E}_a = -\frac{ka}{\varepsilon_0 r^2}\hat{r}$$

$$\vec{E}_\rho = \frac{\hat{r}}{4\pi\varepsilon_0 r^2}\int_a^b \rho_b \cdot 4\pi r^2 dr = \frac{-4\pi k}{4\pi\varepsilon_0 r^2}(b-a)\hat{r}$$

$$\vec{E}_b = +\frac{\left(\frac{k}{b}\right)\cdot 4\pi b^2}{4\pi\varepsilon_0 r^2}\hat{r} = \frac{kb}{\varepsilon_0 r^2}\hat{r}$$

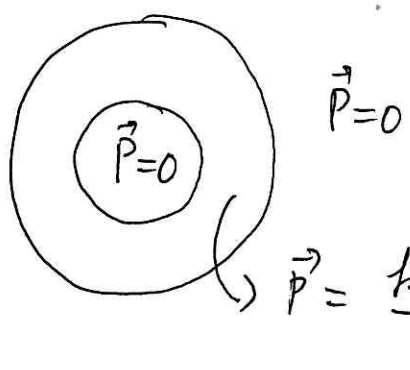
$$\Rightarrow \vec{E}(\vec{r}) = \frac{\hat{r}}{\varepsilon_0 r^2}\left(-ka - k(b-a) + kb\right) = 0.$$

(b)

$$\oint \vec{D}\cdot d\vec{S} = Q_{free;\ enclosed} = 0 \text{ everywhere}$$

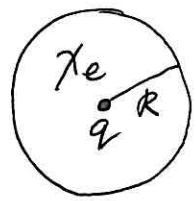
$$\Rightarrow \vec{D}(\vec{r}) = 0 \text{ everywhere.}$$

But $\vec{D} = \varepsilon_0\vec{E} + \vec{P} = 0 \Rightarrow \begin{cases} \vec{E}(r<a) = 0 \\ \vec{E}(a<r<b) = -\frac{\vec{P}}{\varepsilon_0} = -\frac{k\hat{r}}{\varepsilon_0 r} \\ \vec{E}(r>b) = 0 \end{cases}$



just as we found in (a).

P4.32



$$\vec{P} = \varepsilon_0 \chi_e \vec{E} \quad , \vec{D} = \varepsilon \vec{E}$$

$$\varepsilon = \varepsilon_0 (1+\chi_e) \quad , \quad \varepsilon = \varepsilon_0 \varepsilon_r .$$

$$\oint \vec{D} \cdot d\vec{S} = q \Rightarrow D \cdot 4\pi r^2 = q$$

$$\vec{D}(\vec{r}) = \frac{q\,\hat{r}}{4\pi r^2}$$

$$\vec{E} = \frac{1}{\varepsilon}\vec{D} = \frac{q\,\hat{r}}{4\pi\varepsilon r^2} = \frac{q\,\hat{r}}{4\pi\varepsilon_0(1+\chi_e) r^2}$$

$\vec{P} = \varepsilon_0 \chi_e \vec{E}$ for $r < R$ (inside dielectric medium)

$\vec{P} = 0$ for $r > R$ (free space)

$$\Rightarrow \vec{P} = \begin{cases} \dfrac{\chi_e q\,\hat{r}}{4\pi(1+\chi_e) r^2} & , r < R \\ 0 & , r > R \end{cases}$$

$$\rho_b = -\vec{\nabla} \cdot \vec{P} = -\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 P_r\right) = 0 \text{ for all } r.$$

$$\sigma_b = \vec{P} \cdot \hat{n}\Big|_{r=R} = \vec{P} \cdot \hat{r}\Big|_{r=R} = \frac{q\,\chi_e}{4\pi(1+\chi_e) R^2}$$

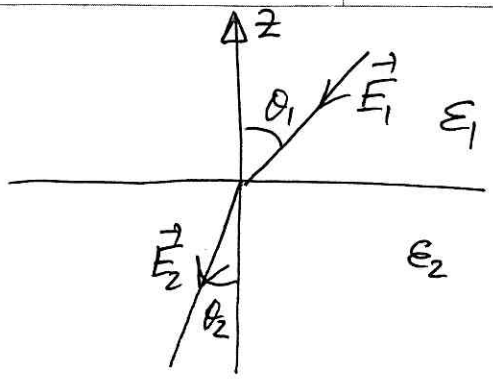
Total bound charge on the outer surface

$$q_b(R) = 4\pi R^2 \cdot \sigma_b = \frac{q\chi_e}{1+\chi_e}$$

$\Rightarrow$ the compensating negative bound charge is right around the point charge, at $r=0$:

$$q_b(r=0) = -\frac{q\chi_e}{1+\chi_e}\,\delta(\vec{r})$$

P 4.33



no free charges at the interface

$$\Rightarrow D_\perp(z=0+) = D_\perp(z=0-)$$

$$\Rightarrow \varepsilon_1 E_1 \cos\theta_1 = \varepsilon_2 E_2 \cos\theta_2 \quad (1)$$

$\vec{\nabla} \times \vec{E} = 0$ always $\Rightarrow E_1^{\parallel} = E_2^{\parallel}$

$$\Rightarrow E_1 \sin\theta_1 = E_2 \sin\theta_2 \quad (2)$$

$$\Rightarrow E_1 = E_2 \frac{\sin\theta_2}{\sin\theta_1}$$

Plug in (1): $\varepsilon_1 \cancel{E_2} \frac{\sin\theta_2}{\sin\theta_1} \cos\theta_1 = \varepsilon_2 \cancel{E_2} \cos\theta_2$

$$\frac{\varepsilon_1}{\tan\theta_1} = \frac{\varepsilon_2}{\tan\theta_2}$$

$$\Rightarrow \frac{\tan\theta_2}{\tan\theta_1} = \frac{\varepsilon_2}{\varepsilon_1}$$

Observe that for $\varepsilon_2 > \varepsilon_1$ $\tan\theta_2 > \tan\theta_1 \Rightarrow \theta_2 > \theta_1$
(for angles $\theta \leq \frac{\pi}{2}$)

$\Rightarrow$ ~~[illegible]~~ convex lens of dielectric with $\varepsilon > 1$ will "defocus" the electric field instead of "focusing" it.